EXPERIMENT 7

DATA VISUALIZATION USING TABLEAU

Aim

To connect, analyze, visualize, and create interactive dashboards using Tableau for effective employee salary analysis and business intelligence.

Software Requirements

• Tableau Desktop / Tableau Public
• Microsoft Excel
• CSV Dataset
• Windows Operating System

Theory

Tableau is a powerful Business Intelligence and Data Visualization tool used to create interactive dashboards and reports. It allows users to connect data sources, create worksheets, build dashboards, create stories, perform data analysis, and share visual insights.

Tableau Architecture

Data Source ↓ Data Connection ↓ Worksheet ↓ Dashboard ↓ Story

Learning Outcomes

✓ Connect employee salary datasets
✓ Create worksheets
✓ Generate salary visualizations
✓ Build dashboards
✓ Create stories
✓ Use filters and parameters
✓ Analyze salary data interactively

Dataset – Employee Salary Analysis Dataset

The dataset contains employee-related information such as Employee ID, Employee Name, Department, Job Role, Education, Experience, Age, Location and Salary. It can be used to compare salaries across departments, roles, experience levels and locations.

Algorithm 1 – Connecting Data Source

Step 1 — Open Tableau Desktop.

Step 2 — Select Connect.

Step 3 — Choose Excel File / Text CSV.

Step 4 — Browse and select the Employee Salary dataset.

Step 5 — Open the dataset.

Step 6 — Verify the imported data.

Procedure

Tableau ↓ Connect ↓ Microsoft Excel / Text CSV ↓ Employee Salary Dataset ↓ Open

Output

Employee Salary dataset loaded into Tableau successfully.

Algorithm 2 – Creating Worksheet

Step 1 — Open Sheet 1.

Step 2 — Drag Department to Columns.

Step 3 — Drag Salary to Rows.

Step 4 — Set Salary aggregation to Average.

Step 5 — Select an appropriate visualization.

Step 6 — Analyze the results.

Output Visualization

Department-wise Average Salary – The visualization compares the average salary of employees across departments.

Observation: The chart helps identify departments with comparatively higher and lower average salaries.

Algorithm 3 – Bar Chart Creation

Step 1 — Drag Job Role → Columns.

Step 2 — Drag Salary → Rows.

Step 3 — Select Bar Chart.

Step 4 — Apply data labels.

Step 5 — Format the visualization.

Output Visualization

Employee Salary Comparison – Comparison of salaries across different job roles.

Interpretation: The bar chart makes it easy to identify job roles associated with higher or lower average salaries.

Algorithm 4 – Pie Chart Creation

Step 1 — Drag Department.

Step 2 — Drag Salary.

Step 3 — Select Pie Chart.

Step 4 — Show Labels.

Step 5 — Analyze Salary Distribution.

Output Visualization

Department-wise Salary Contribution – Contribution of different departments to the total recorded salary.

Interpretation: The visualization helps compare the salary contribution of each department.

Algorithm 5 – Line Chart Creation

Step 1 — Drag Experience.

Step 2 — Drag Salary.

Step 3 — Select Line Chart.

Step 4 — Apply formatting.

Step 5 — Analyze the salary trend.

Output Visualization

Experience vs Salary Trend – Shows how salary changes with employee experience.

Algorithm 6 – Creating Calculated Field

Objective: Calculate Salary Category.

Example Tableau expression: IF [Salary] >= 1000000 THEN "High Salary" ELSEIF [Salary] >= 500000 THEN "Medium Salary" ELSE "Low Salary" END

Procedure: Analysis → Create Calculated Field → Name: Salary Category → Enter formula → Apply.

Output Visualization

Salary categories can be used to group employees into low, medium and high salary ranges for further analysis.

Algorithm 7 – Applying Filters

Step 1 — Drag Department to Filters.

Step 2 — Select a Department.

Step 3 — Apply Filter.

Step 4 — Observe the updated charts.

The selected department filter updates the salary charts and dashboard to show only the relevant employee records.

Algorithm 8 – Creating Parameter

Step 1 — Create a Parameter.

Step 2 — Specify Minimum Salary.

Step 3 — Apply the Parameter.

Step 4 — Display matching employee records.

Example: Minimum Salary = 500000

The parameter displays only records satisfying Salary ≥ 500000.

Algorithm 9 – Dashboard Creation

Step 1 — Create multiple worksheets.

Step 2 — Select Dashboard.

Step 3 — Drag worksheets into the dashboard.

Step 4 — Arrange components.

Step 5 — Add filters and parameters.

Step 6 — Save the dashboard.

Dashboard Layout

Employee Salary Analysis Dashboard → KPI Cards → Bar Chart + Pie Chart + Line Chart → Filter Panel

Dashboard Analysis

KPI	Value / Measure
Total Employees	COUNTROWS(EmployeeSalary)
Average Salary	AVERAGE(Salary)
Highest Salary	MAX(Salary)
Lowest Salary	MIN(Salary)

KPI cards provide a quick summary of employee count and salary statistics, while charts provide department-wise, role-wise and experience-based comparisons. Filters allow interactive exploration.

Algorithm 10 – Story Creation

Step 1 — Create Dashboard.

Step 2 — Select Story.

Step 3 — Add Dashboard.

Step 4 — Add Story Points.

Step 5 — Present Insights.

Example Story

Story Point 1 – Overall Salary Analysis
Story Point 2 – Department-wise Salary Comparison

Story Point 3 – Job Role and Experience Analysis
Story Point 4 – Final Conclusions

Applications

Domain — Application
Human Resources — Employee Compensation Analysis
Finance — Salary Budget Planning
Management — Workforce Cost Analysis
Business Analytics — Compensation Dashboard

Result

The Employee Salary dataset was successfully connected and analyzed using Tableau. Worksheets and multiple visualizations including bar charts, pie charts and line charts were generated. A calculated field, filters and parameters were applied for interactive analysis. Finally, the visualizations were combined into a dashboard and organized into a story.

Conclusion

Tableau provides an effective platform for transforming employee salary data into interactive visual insights. This experiment demonstrates the workflow from connecting a salary dataset to creating worksheets, applying analytical controls, designing a dashboard and presenting findings through a story.